

TrES-1b

R-0239

Benchmark run · workflow W8-benchmark-deep · record deep-bench_TRES-1_250913 · created 2026-07-17T04:53:01+00:00

PRACTICE / VALIDATION RUN — not submitted to any science database

Results

Mid-Transit Time (Tmid)	2460931.6705 +/- 0.0032 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.184 +/- 0.052
Transit depth (Rp/Rs)^2	3.37 +/- 1.92 [%]
Orbital Inclination (inc)	86.63 +/- 1.47
Airmass coefficient 1 (a1)	0.6 +/- 0.26
Airmass coefficient 2 (a2)	0.37 +/- 0.66
Scatter in the residuals of the lightcurve fit is	44.63 %
Best Comparison Star	#2 - [590, 82]
Optimal Aperture	1.34
Optimal Annulus	3.5
Transit Duration (day)	0.084 +/- 0.021

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.184 ± 0.052 vs 0.1358 (deviation 0.93σ)
Duration fit vs published	0.084 d vs 0.1042 d (deviation 0.96σ)
Mid-time O–C	7.2 min
Depth SNR	3.5
Residual scatter	44.63 %

Light curve

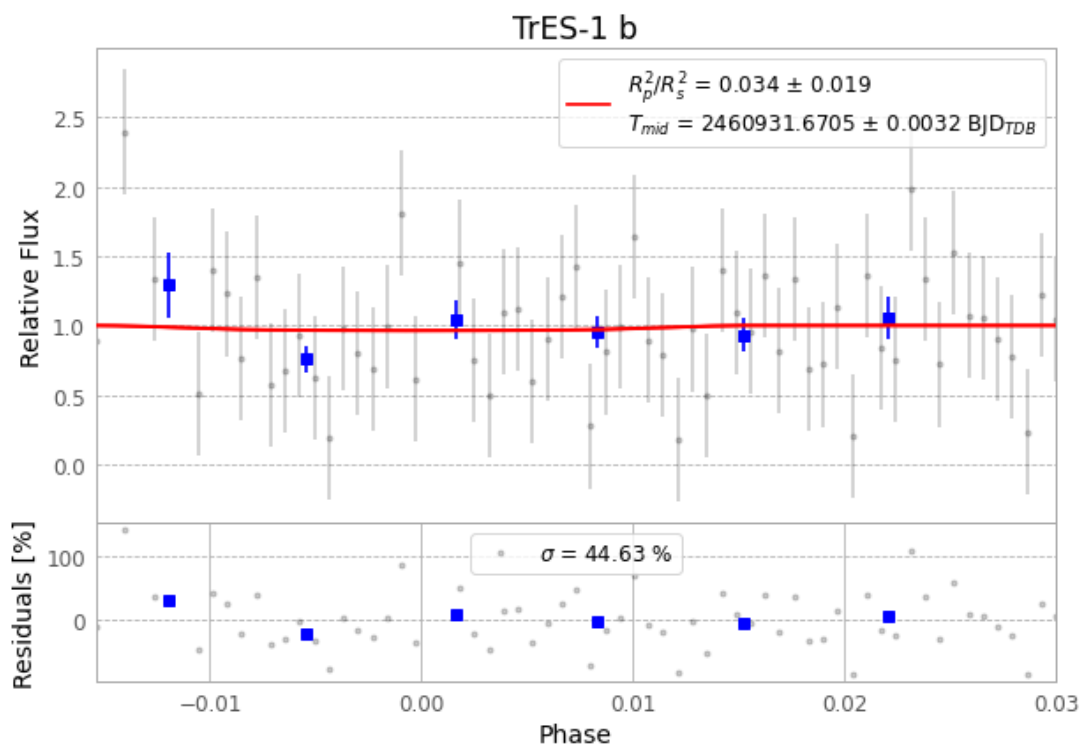


Figure 1 — FinalLightCurve_TrES-1 b_2025-09-12.png (SHA-256 0d5d8b405b1d669a...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [531, 149], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 7363d9c11497ed26...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit 825eb7e

Data provenance

66 FITS frames (43 MB), TRES-1250913025413.FITS → TRES-1250913061216.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-TRES-1-250913.log (99b4c1354143...), FinalLightCurve_TrES-1 b_2025-09-12.png (0d5d8b405b1d...), FinalLightCurve_TrES-1 b_2025-09-12.csv (096900cc57ef...)

Compute resources

Wall time 160.3 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-17 04:53Z.